DPP 07

PHYSICS · Motion in a Straight Line

NEET 2027 · Daily Practice Problems

Time: 20 min
Marks: +4 / −1

Each question has four options, of which only one is correct.

1 A particle moves along the x-axis as $x = 2t^2 - 4t + 3$, where x is in metres and t in seconds. Its velocity at t = 3 s is

(1) 4 m/s (2) 8 m/s
(3) 12 m/s (4) 6 m/s

2 A car starts from rest and accelerates uniformly at 2 m/s² for 5 s. The distance it covers in this time is

(1) 25 m (2) 10 m
(3) 50 m (4) 12.5 m

3 A ball is dropped from a height of 80 m. Taking g = 10 m/s², the time it takes to reach the ground is

(1) 2 s (2) 8 s
(3) 4 s (4) 16 s

4 A body covers the first half of a distance at 20 km/h and the second half at 30 km/h. Its average speed for the whole journey is

(1) 24 km/h (2) 25 km/h
(3) 26 km/h (4) 50 km/h

5 The slope of a velocity–time graph gives

(1) displacement (2) speed
(3) acceleration (4) distance

6 A ball is thrown vertically upward at 20 m/s. Taking g = 10 m/s², the maximum height it reaches is

(1) 10 m (2) 40 m
(3) 30 m (4) 20 m

7 A train 150 m long moves at 54 km/h. The time it takes to cross a bridge 300 m long is

(1) 20 s (2) 30 s
(3) 10 s (4) 45 s

8 A body starts from rest with a uniform acceleration of 4 m/s². The distance it covers in the 3rd second is

(1) 10 m (2) 18 m
(3) 8 m (4) 12 m

9 The area under an acceleration–time graph gives

(1) change in velocity (2) displacement

(3) distance (4) acceleration

10 Two cars move in the same direction at 60 km/h and 40 km/h. The velocity of the faster car relative to the slower one is

(1) 100 km/h (2) 40 km/h

(3) 60 km/h (4) 20 km/h

11 The velocity of a particle is $v = 3t^2$ m/s. Its acceleration at $t = 2$ s is

(1) 6 m/s² (2) 24 m/s²

(3) 12 m/s² (4) 3 m/s²

12 A stone is dropped from rest. The ratio of the distances it falls in the 1st, 2nd and 3rd seconds is

(1) 1 : 2 : 3 (2) 1 : 4 : 9

(3) 1 : 3 : 5 (4) 1 : 1 : 1

Space for rough work

KEY

ANSWER KEY · DPP 07

Physics · Motion in a Straight Line

Q.	1	2	3	4	5	6
Ans.	(2)	(1)	(3)	(1)	(3)	(4)
Q.	7	8	9	10	11	12
Ans.	(2)	(1)	(1)	(4)	(3)	(3)

Hints

1. $v = dx/dt = 4t - 4 \rightarrow 8$ m/s

2. $s = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 25 = 25$ m

3. $t = \sqrt{(2h/g)} = \sqrt{16} = 4$ s

4. $\bar{v} = 2v_1v_2 / (v_1 + v_2) = 24$ km/h

6. $H = u^2/2g = 400/20 = 20$ m

7. $t = (150 + 300) / 15 = 30$ s

8. $s_n = u + a(2n - 1)/2 = 10$ m

11. $a = dv/dt = 6t = 12$ m/s²